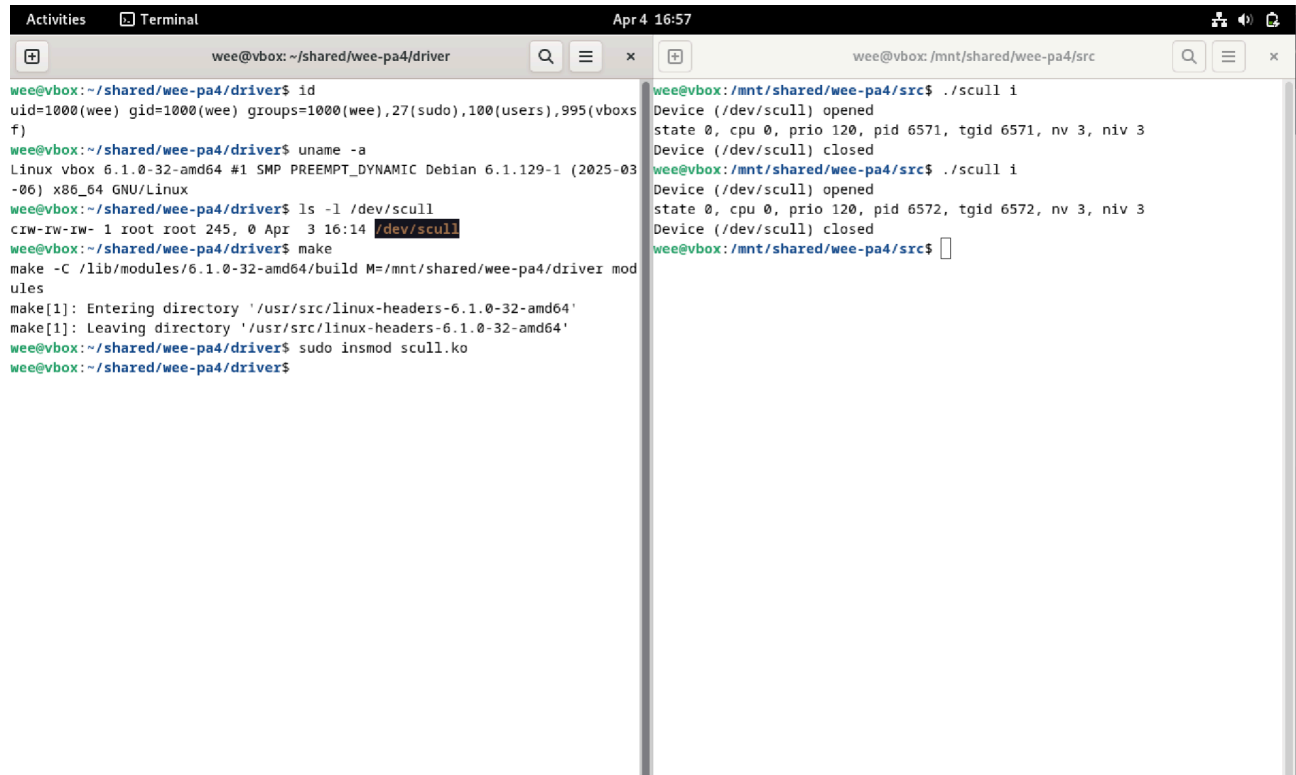


William Ee

Due 4/4/2025

I pledge my honor that I have abided by the Stevens Honor System.

3.2 - Test runs of ./scull i user-space test command



```
Activities Terminal Apr 4 16:57
wee@vbox: ~/shared/wee-pa4/driver
wee@vbox:~/shared/wee-pa4/driver$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa4/driver$ uname -a
Linux vbox 6.1.0-32-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.129-1 (2025-03-06) x86_64 GNU/Linux
wee@vbox:~/shared/wee-pa4/driver$ ls -l /dev/scull
crw-rw-rw- 1 root root 245, 0 Apr  3 16:14 /dev/scull
wee@vbox:~/shared/wee-pa4/driver$ make
make -C /lib/modules/6.1.0-32-amd64/build M=/mnt/shared/wee-pa4/driver modules
make[1]: Entering directory '/usr/src/linux-headers-6.1.0-32-amd64'
make[1]: Leaving directory '/usr/src/linux-headers-6.1.0-32-amd64'
wee@vbox:~/shared/wee-pa4/driver$ sudo insmod scull.ko
wee@vbox:~/shared/wee-pa4/driver$

wee@vbox: /mnt/shared/wee-pa4/src$ ./scull i
Device (/dev/scull) opened
state 0, cpu 0, prio 120, pid 6571, tgid 6571, nv 3, niv 3
Device (/dev/scull) closed
wee@vbox: /mnt/shared/wee-pa4/src$ ./scull i
Device (/dev/scull) opened
state 0, cpu 0, prio 120, pid 6572, tgid 6572, nv 3, niv 3
Device (/dev/scull) closed
wee@vbox: /mnt/shared/wee-pa4/src$
```

4.1 - Extending with log information

```
wee@vbox: ~/shared/wee-pa4/driver

wee@vbox:~/shared/wee-pa4/driver$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa4/driver$ sudo insmod scull.ko
wee@vbox:~/shared/wee-pa4/driver$ ../src/scull i
Device (/dev/scull) opened
state 0, cpu 0, prio 120, pid 6893, tgid 6893, nv 5, niv 2
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa4/driver$ ../src/scull i
Device (/dev/scull) opened
state 0, cpu 0, prio 120, pid 6895, tgid 6895, nv 3, niv 4
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa4/driver$ sudo rmmod scull
wee@vbox:~/shared/wee-pa4/driver$ sudo dmesg | tail -6
[41532.382964] scull open
[41532.383025] scull close
[41541.647361] scull open
[41541.647430] scull close
[41547.323519] Task: PID 6893, TGID 6893
[41547.323521] Task: PID 6895, TGID 6895
wee@vbox:~/shared/wee-pa4/driver$
```

4.2 - Test run of ./scull p user-space test command

```
wee@vbox: ~/shared/wee-pa4/driver

wee@vbox:~/shared/wee-pa4/driver$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa4/driver$ sudo insmod scull.ko
wee@vbox:~/shared/wee-pa4/driver$ ../src/scull p
Device (/dev/scull) opened
state 0, cpu 0, prio 120, pid 7426, tgid 7426, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7426, tgid 7426, nv 0, niv 2
state 0, cpu 0, prio 120, pid 7425, tgid 7425, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7425, tgid 7425, nv 0, niv 2
state 0, cpu 0, prio 120, pid 7424, tgid 7424, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7424, tgid 7424, nv 0, niv 2
state 0, cpu 0, prio 120, pid 7423, tgid 7423, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7423, tgid 7423, nv 0, niv 2
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa4/driver$ sudo rmmod scull
wee@vbox:~/shared/wee-pa4/driver$ sudo dmesg | tail -6
[44792.065152] scull open
[44792.065734] scull close
[44802.780104] Task 1: PID 7426, TGID 7426
[44802.780108] Task 2: PID 7425, TGID 7425
[44802.780108] Task 3: PID 7424, TGID 7424
[44802.780109] Task 4: PID 7423, TGID 7423
wee@vbox:~/shared/wee-pa4/driver$
```

4.3 - Test run of ./scull t user-space test command

```
wee@vbox: ~/shared/wee-pa4/driver
wee@vbox:~/shared/wee-pa4/driver$ id
uid=1000(wee) gid=1000(wee) groups=1000(wee),27(sudo),100(users),995(vboxsf)
wee@vbox:~/shared/wee-pa4/driver$ sudo insmod scull.ko
wee@vbox:~/shared/wee-pa4/driver$ ../src/scull t
Device (/dev/scull) opened
state 0, cpu 0, prio 120, pid 7447, tgid 7443, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7447, tgid 7443, nv 0, niv 2
state 0, cpu 0, prio 120, pid 7446, tgid 7443, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7446, tgid 7443, nv 0, niv 2
state 0, cpu 0, prio 120, pid 7445, tgid 7443, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7445, tgid 7443, nv 0, niv 2
state 0, cpu 0, prio 120, pid 7444, tgid 7443, nv 0, niv 0
state 0, cpu 0, prio 120, pid 7444, tgid 7443, nv 0, niv 2
Device (/dev/scull) closed
wee@vbox:~/shared/wee-pa4/driver$ sudo rmmod scull
wee@vbox:~/shared/wee-pa4/driver$ sudo dmesg | tail -6
[44928.825293] scull open
[44928.825543] scull close
[44949.311045] Task 1: PID 7447, TGID 7443
[44949.311048] Task 2: PID 7446, TGID 7443
[44949.311049] Task 3: PID 7445, TGID 7443
[44949.311049] Task 4: PID 7444, TGID 7443
wee@vbox:~/shared/wee-pa4/driver$
```